Mohamed Eissa Abdelkareem

AI Engineer · Generative AI & LLMs

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Location: Cairo, Egypt
Military Status: Completed

Profile

Computer Science graduate (2024) building **Generative AI** applications in **Python** with **Large Language Models (LLMs)**: LLM API integration, prompt engineering, and LLM-driven intent detection and intent-to-action mapping. Built a Gemini-powered desk assistant that turns natural-language commands into device responses and actions, and an **Arabic OCR** pipeline that extracts, validates and stores fields from scanned identity documents, with output checked against the source images. Took a client product from requirements to production deployment: a full-stack booking platform on **PostgreSQL** with online payments and row-level security. Strong in Python, REST API integration, SQL, Linux and Git; comfortable owning a component end to end, and currently building workflow automations with **n8n**.

Core Competencies & Technical Skills

- **Programming:**
  - ➢ **Python**, **Java**, **C/C++** (fundamentals), **SQL**, Object-Oriented Programming (**OOP**).
- **Generative AI & LLMs:**
  - ➢ **LLM API integration** (**Gemini**, **OpenAI**), **Prompt Engineering**, **intent detection** and intent-to-action mapping with LLMs, model/tool selection and comparison, evaluating model output against ground truth.
- **Document Intelligence & Vision:**
  - ➢ **OCR** (**Tesseract**), Arabic & English document data extraction, image pre-processing (**OpenCV**), field parsing and rule-based validation.
- **Automation & Scripting:**
  - ➢ **n8n** workflow automation (personal projects), **Python** automation scripts for data collection into **Excel**.
- **Backend & Data:**
  - ➢ **REST API** integration (Gemini, Paymob), relational database design (**MySQL**, **PostgreSQL**), **Supabase**, **Postman**.
- **Tools & Deployment:**
  - ➢ **Git & GitHub**, **Linux**, **Vercel**, **ESP32-S3** / **Arduino** (hardware–software integration).

Education

- **Bachelor of Computer Science** Future University in Egypt (FUE) *Graduated: 2024*
  - ➢ Degree awarded in academic affiliation with the **University of Cincinnati** (USA).
- **Graduation Project:** Multi-Model Authentication System.
  - ➢ *Brief:* Built the **Arabic OCR** and data-extraction module of a multi-model authentication system.

Key Technical Projects

- **Wanas — LLM-Driven Conversational Desk Assistant (Independent Project, 2026):**
  - ➢ Built an interactive desk robot on an **ESP32-S3** microcontroller that responds to natural-language commands.
  - ➢ Used **Gemini LLM APIs** over **REST** for **intent detection** on natural-language commands, mapping each detected intent to a device response or action.
  - ➢ Handled sensor input, Wi-Fi communication and response latency under embedded memory and timing constraints.
  - ➢ **Stack:** Python, C/C++, ESP32-S3, REST APIs, Gemini API.

- **Document Data Extraction Pipeline — Arabic OCR (Graduation Project, 2024):**
  - ➢ Built the **OCR** and data-extraction component of a multi-model authentication system.
  - ➢ Processed scanned Egyptian national ID cards end to end: **image pre-processing**, **Arabic-script field recognition**, parsing into structured records, and persistence to **MySQL**.
  - ➢ Tested extracted fields against the source images and handled failure cases (low-quality scans, partial fields) before storage.
  - ➢ **Stack:** Python, Tesseract OCR (pytesseract), OpenCV, MySQL.
- **Shadia Travel — Tourism Platform, Mobile & Web (Freelance, 2026):**
  - ➢ Owned a client product from requirements to production: mobile client, admin dashboard and public web portal in one monorepo.
  - ➢ Built booking and online payment flows end to end (**Paymob**) on a **Supabase/PostgreSQL** backend secured with **row-level security**; deployed on **Vercel**.
  - ➢ **Stack:** Flutter (Dart), Riverpod, Next.js (React, TypeScript), Tailwind CSS, Supabase (PostgreSQL, RLS), Paymob, Vercel.
- **Workflow Automation — n8n & Python (Personal Projects, Ongoing):**
  - ➢ Building **n8n** workflows that automate repetitive day-to-day tasks by connecting apps and services.
  - ➢ Wrote a **Python** script that collects data automatically and writes it into structured **Excel** sheets, replacing manual copy-paste.
  - ➢ **Stack:** n8n, Python, Microsoft Excel.
- **Gate Event System (Python Automation):**
  - ➢ Developed a desktop tool using **Python (Tkinter)** to automate event entry logging and record management.
- **PlayStation Center Management System:**
  - ➢ Developed a time-based billing system using **Android** and **Java**, with real-time session tracking and automated cost calculation.

Technical Experience & Internships

Freelance Developer *Freelance | Cairo*
Mar 2026 – Present

- Work directly with clients to turn requirements and use cases into working applications and **AI-integrated projects**, from scoping through delivery.
- Delivered the **Shadia Travel** platform (booking, payments, admin dashboard) to production.

IT Support Intern *Bayan Tech | Cairo*
Apr 2026 – Jun 2026

- Provided hardware, software and end-user support across the office.

IT Help Desk (Military Service) *Military Engineers Department – Systems Branch | Cairo*
Mar 2025 – Mar 2026

- Designed and implemented a **digital archive and tracking system** for administrative documents and technical estimations, replacing a manual paper-based process.
- Maintained branch network infrastructure, troubleshot technical issues and supported users across the department.

Leadership & Communication

- Head of PR – Google Developer Student Clubs (**GDSC**) FUE.

Language Skills

- **Arabic**: Native.
- **English**: Professional working proficiency.

References

- Available on request